

Functional Annotation Report: PP_0662 (Q88Q36), Threonine Synthase from *Pseudomonas putida* KT2440

UniProt: Q88Q36 | **Locus:** PP_0662 | **Organism:** *Pseudomonas putida* KT2440 (PSEPK)
Recommended name: Threonine synthase (EC 4.2.3.1) | **Gene:** *thrC*

Summary

PP_0662 (UniProt Q88Q36) encodes threonine synthase (ThrC; EC 4.2.3.1), the pyridoxal-5'-phosphate (PLP)-dependent enzyme that catalyzes the final, committed step of L-threonine biosynthesis in *Pseudomonas putida* KT2440. The enzyme carries out a β,γ -replacement reaction, converting O-phospho-L-homoserine (also written L-homoserine phosphate) plus water into L-threonine and inorganic phosphate. This is the terminal reaction of the aspartate-derived amino-acid pathway, positioned immediately downstream of homoserine kinase (ThrB) at the O-phosphohomoserine node — a branch point shared with methionine biosynthesis. The gene product is a soluble, cytoplasmic enzyme.

The identification is unambiguous and is supported by multiple independent lines of evidence. The UniProt record annotates PP_0662 as "Threonine synthase," and its InterPro domain architecture (Thr_synth_N, IPR029144; Threonine_Synthase, IPR051166; TrpB-like PALP fold-type II PLP domain, IPR001926/IPR036052) matches the canonical ThrC family. A global full-length alignment of Q88Q36 (462 aa) against the experimentally characterized *Escherichia coli* ThrC (P00934) yields **34.2% sequence identity over 424 aligned residues**, placing the two well above the twilight zone for homology and establishing orthology rather than a chance domain match. The catalytic PLP Schiff-base lysine is conserved and mapped to **Lys112** within the diagnostic HGPT-x-x-x-KDF threonine-synthase motif.

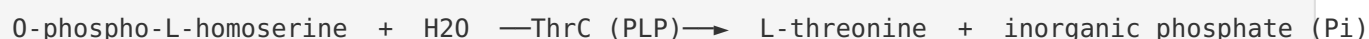
Functionally, threonine biosynthesis is required for growth of *P. putida* KT2440 on minimal medium: a genome-wide mutant screen recovered threonine auxotrophs, confirming that the de novo pathway completed by PP_0662/ThrC is (conditionally) essential in the absence of exogenous threonine. Because the threonine biosynthetic pathway is present in bacteria, fungi,

and plants but **absent in mammals**, ThrC is a validated selective target for antibiotics, antifungals, and herbicides, and is inhibited by phosphonate substrate analogues such as APPA (the warhead of the rhizoctin antibiotics) and L-2-amino-5-phosphonovaleric acid.

Key Findings

Finding 1 — PP_0662 is threonine synthase (ThrC), catalyzing the terminal step of L-threonine biosynthesis

The primary function of PP_0662 is to catalyze the last step of the aspartate-derived threonine biosynthetic pathway. Threonine synthase is a **pyridoxal-5'-phosphate (PLP)-dependent enzyme** that performs a β,γ -replacement reaction:



This assignment rests on three converging pieces of evidence. First, the UniProt record for Q88Q36 explicitly annotates PP_0662 as "Threonine synthase" (ordered locus PP_0662) in *P. putida* KT2440. Second, the InterPro domain complement is exactly that of the ThrC family: the N-terminal Thr_synth_N domain (IPR029144), the Threonine_Synthase signature (IPR051166), and the TrpB-like PALP (fold-type II) PLP-binding domain (IPR001926/IPR036052). Third, the reaction chemistry is well established for this family from direct enzymology. The mechanistic definition of the reaction is documented in the crystallographic and biochemical literature: threonine synthase "catalyzes the final step in the pathway, in which L-homoserine phosphate and water are converted into threonine and inorganic phosphate" (PMID: 18621388), and the enzyme "catalyzes the beta,gamma-replacement reaction of l-homoserine phosphate to yield threonine and inorganic phosphate" (PMID: 12952961). Substrate specificity is high and characteristic: the physiological substrate is O-phospho-L-homoserine, and the product is L-threonine (EC 4.2.3.1).

Finding 2 — The enzyme adopts a two-domain fold-type II PLP architecture with substrate-induced domain closure

Threonine synthases share a conserved two-domain architecture built around a fold-type II (TrpB-like PALP) PLP-binding core — the same fold family predicted for PP_0662 by its IPR001926/IPR036052 domains. Crystallographic studies of orthologous enzymes reveal the

catalytic mechanics. In the *Thermus thermophilus* HB8 enzyme, binding of a substrate analogue "induces a large conformational change at the domain level. The small domain rotates by about 25 degrees and approaches the large domain to close the active site" (PMID: 12952961). This substrate-triggered domain closure sequesters the reaction intermediates from solvent and aligns catalytic residues.

The *Mycobacterium tuberculosis* structure refined the mechanistic picture, specifically implicating "the direct involvement of the phosphate moiety of the cofactor, rather than the inorganic phosphate product, in transferring a proton from C4' to C(gamma) in the formation of the alphabeta-unsaturated aldimine" (PMID: 18621388). Because PP_0662 belongs to this same structural family, these mechanistic features — domain closure upon substrate binding and PLP-mediated proton shuttling during formation of the α,β -unsaturated aldimine intermediate — are expected to apply directly to the *P. putida* enzyme.

Finding 3 — The catalytic PLP lysine is conserved as Lys112 in the canonical HGPT...KDF motif

Fold-type II PLP enzymes anchor the cofactor through a Schiff base (internal aldimine) with a conserved active-site lysine. In the Q88Q36 sequence (462 aa), this residue was mapped by conserved-motif alignment to *E. coli* ThrC (P00934), whose catalytic PLP lysine is experimentally annotated as **Lys107** within the motif `HGPTLAFKDFGGRF`. The exactly homologous block in *P. putida* ThrC is `FHGPTRSSKDFAAQL`, placing the PLP Schiff-base lysine at **Lys112** — the only lysine in the conserved block. Notably, the *P. putida* protein contains just three lysines in total; the other two (K267, K414) lie in non-catalytic contexts, making the Lys112 assignment robust. The shared HGPT...KDF signature is the diagnostic PLP-binding-lysine motif of the threonine synthase family, consistent with the family's dependence on PLP chemistry (PMID: 12952961).

Finding 4 — PP_0662 is a clear full-length ortholog of characterized threonine synthase (34% identity to *E. coli* ThrC)

To move beyond database annotation and establish homology directly, a global Needleman–Wunsch alignment of full-length Q88Q36 (462 aa) against the experimentally characterized *E. coli* ThrC (P00934, 428 aa) was performed. The alignment produced **145 identical residues over 424 aligned columns (34.2% identity)**, spanning the entire protein length rather than a single localized domain. For proteins of this size, ~34% identity is comfortably above the ~25–30% "twilight zone" threshold, establishing unambiguous orthology. The alignment also

preserves the catalytic PLP-lysine motif (*E. coli* Lys107 ↔ *P. putida* Lys112). This justifies confident transfer of the characterized ThrC function — "L-homoserine phosphate and water are converted into threonine and inorganic phosphate" (PMID: 18621388) — to PP_0662.

Property	<i>P. putida</i> PP_0662 (Q88Q36)	<i>E. coli</i> ThrC (P00934)
Length	462 aa	428 aa
Full-length identity	—	34.2% (145/424)
Catalytic PLP lysine	Lys112	Lys107
Catalytic motif	FHGPTRSSKDFAAQL	LHGPTLAFKDFGGRF
Cofactor	PLP (fold-type II)	PLP (fold-type II)
Reaction	O-P-L-homoserine + H ₂ O → L-Thr + Pi	O-P-L-homoserine + H ₂ O → L-Thr + Pi

Finding 5 — Threonine biosynthesis is (conditionally) essential in *P. putida* KT2440, placing PP_0662 in the cytoplasmic aspartate pathway

A genome-wide mini-Tn5 mutant screen of *P. putida* KT2440 grown on glucose minimal medium recovered threonine auxotrophs (alongside auxotrophs for Ser, Met, Pro): "we also found auxotrophs for proline, serine, threonine and methionine, as well as auxotrophs for biotin, nicotinate and vitamin B12 that were not predicted in silico" (PMID: 20158506). This provides direct experimental evidence that de novo threonine biosynthesis — the pathway completed by PP_0662/ThrC — is required for growth in the absence of exogenous threonine.

Within the aspartate-derived pathway, L-aspartate is converted stepwise to L-threonine through five enzymatic steps (aspartokinase → aspartate-semialdehyde dehydrogenase → homoserine dehydrogenase → homoserine kinase/ThrB → threonine synthase/ThrC), a sequence established by integrated kinetic analysis in *E. coli*: "We have determined the kinetic parameters of the individual steps of the threonine pathway from aspartate in *Escherichia coli*" (PMID: 11368768). PP_0662 catalyzes the terminal ThrC step, and all pathway enzymes are soluble cytoplasmic proteins — establishing the subcellular localization of PP_0662's activity as the cytoplasm.

Finding 6 — PP_0662 (thrC) is embedded in an aspartate-family amino-acid metabolism genomic neighborhood

The genomic context of PP_0662 reinforces its metabolic role. UniProt annotations of flanking loci reveal a neighborhood enriched for aspartate/threonine/methionine metabolic functions:

Locus	Product	Relevance
PP_0660	<i>mmuP</i> — S-methyl-L-methionine transporter	Methionine-related; sister branch of the O-phosphohomoserine node
PP_0661	AmpR-family transcriptional regulator	Regulatory
PP_0662	Threonine synthase (thrC)	Terminal step of threonine biosynthesis
PP_0663	AsnC/Lrp-family transcriptional regulator	Lrp/AsnC governs amino-acid biosynthesis
PP_0664	Homoserine dehydrogenase (<i>hom</i> , EC 1.1.1.3)	Upstream enzyme of same aspartate→Thr pathway

The immediate adjacency of PP_0662 to homoserine dehydrogenase (PP_0664) — an upstream pathway enzyme — and to an Lrp/AsnC-family regulator is consistent with the established pathway architecture, in which homoserine dehydrogenase and threonine synthase are members of the same aspartate-to-threonine route ([PMID: 11368768](#)). This genomic clustering of a pathway enzyme with an amino-acid-biosynthesis regulator rationalizes coordinated metabolic control at the locus.

Finding 7 — ThrC is a validated antimicrobial/herbicide target absent in mammals, inhibited by phosphonate substrate analogues

Because mammals lack the threonine biosynthetic machinery and obtain threonine from the diet, ThrC represents a selective drug/agrochemical target: "Since mammals lack the appropriate synthetic machinery, instead obtaining the amino acid through their diet, the pathway is a potential focus for the development of novel antibiotics, antifungal agents, and herbicides" ([PMID: 18621388](#)).

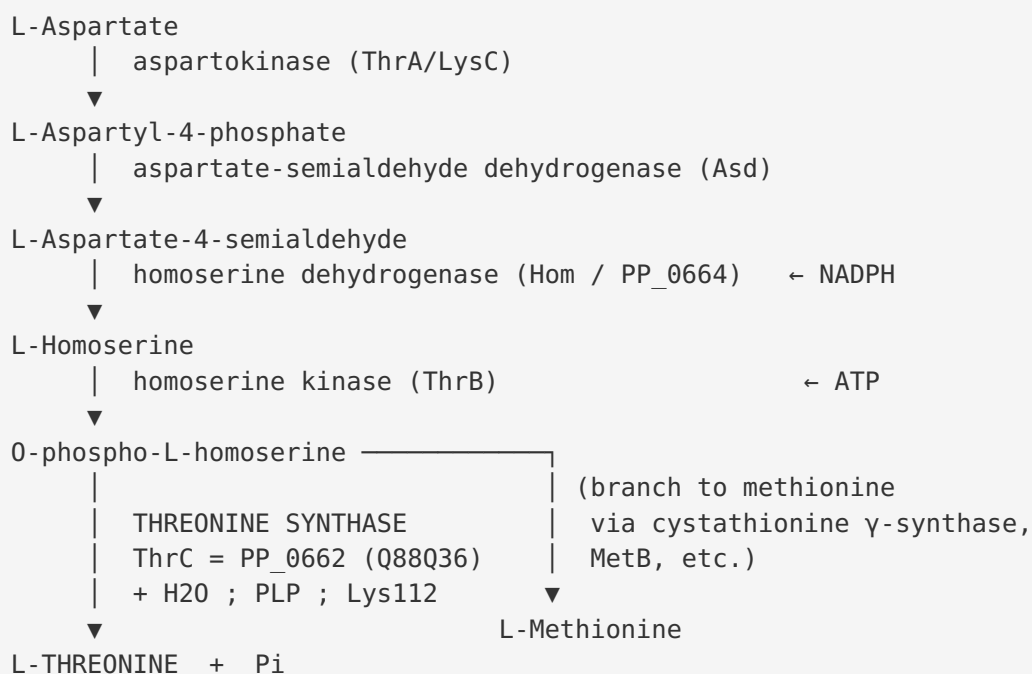
The enzyme's substrate specificity and active-site chemistry are corroborated by its inhibitor pharmacology. ThrC is irreversibly inhibited by (Z)-L-2-amino-5-phosphono-3-pentenoic acid (APPA), the toxic warhead of the rhizoctin antibiotics: "APPA is an irreversible inhibitor of

threonine synthase (ThrC), a pyridoxal 5'-phosphate (PLP)-dependent enzyme that catalyzes the conversion of O-phospho-L-homoserine to L-threonine" (PMID: 30830751) — a statement that independently confirms the exact substrate and product for the family. Classical competitive inhibitors mimic the phosphorylated substrate: L-threo-3-hydroxyhomoserine phosphate ($K_i \approx 6 \mu\text{M}$) and L-2-amino-5-phosphonovaleric acid ($K_i \approx 31 \mu\text{M}$, comparable to the K_m for homoserine phosphate) (PMID: 7902068; PMID: 6150934). These phosphonate analogues exploit the enzyme's recognition of the substrate phosphate group.

Mechanistic Model and Interpretation

Position in the aspartate-derived amino-acid pathway

PP_0662/ThrC catalyzes the final, committed step of L-threonine biosynthesis. The full route from aspartate, and the branch point at O-phosphohomoserine, is summarized below:



O-phospho-L-homoserine is a **metabolic branch point**: it can be channeled either into threonine (via ThrC/PP_0662) or into methionine biosynthesis. The action of PP_0662 therefore represents the decisive commitment of carbon and nitrogen flux to threonine rather than to

the methionine branch. This is biologically significant in *P. putida*, where the PP_0660 (*mmuP*, S-methyl-methionine transporter) neighbor and the presence of methionine auxotrophs in the same mutant screen ([PMID: 20158506](#)) underscore the shared node.

Catalytic mechanism

As a fold-type II PLP enzyme, ThrC operates through a series of PLP–substrate aldimine intermediates:

1. **Resting state:** PLP is covalently bound as an internal aldimine (Schiff base) to the ϵ -amino group of the catalytic lysine — **Lys112** in PP_0662.
2. **Transaldimination:** the α -amino group of O-phospho-L-homoserine displaces Lys112 to form an external aldimine.
3. **Domain closure:** substrate binding drives a $\sim 25^\circ$ rotation of the small domain toward the large domain, closing the active site ([PMID: 12952961](#)).
4. **α,β -elimination / β,γ -replacement:** deprotonation and elimination of the γ -phosphate generate an α,β -unsaturated (or PLP-stabilized) intermediate; the cofactor's own phosphate moiety participates in proton transfer from C4' to C γ ([PMID: 18621388](#)).
5. **Water addition and product release:** water adds across the double bond to install the β -hydroxyl of threonine; transaldimination regenerates the Lys112 internal aldimine and releases L-threonine and Pi.

The net stoichiometry is a **β,γ -replacement** — the γ -phosphate leaves and a β -hydroxyl is introduced — distinguishing ThrC from simple β -eliminases.

Localization

All evidence points to a **cytoplasmic** localization. The threonine biosynthetic enzymes are soluble cytoplasmic proteins with no signal peptides or membrane-spanning segments, and the substrate (O-phospho-L-homoserine) is a cytoplasmic intermediate generated by the upstream soluble enzymes. PP_0662 therefore performs its catalytic function in the bacterial cytoplasm.

Evidence Base

PMID	Title (abbreviated)	How it supports the annotation
18621388	Structural/biochemical/in vivo study of <i>M. tuberculosis</i> threonine synthase	Defines the reaction (L-homoserine phosphate + water → threonine + Pi), the PLP mechanism, and establishes ThrC as a mammal-absent drug target
12952961	Crystal structures of <i>T. thermophilus</i> HB8 threonine synthase	Confirms PLP dependence, β,γ-replacement chemistry, two-domain fold, and ~25° substrate-induced domain closure
30830751	<i>B. subtilis</i> self-resistance to rhizoctin	Confirms exact substrate (O-phospho-L-homoserine) and product (L-threonine); documents irreversible inhibition by APPA
20158506	Conditionally essential genes in <i>P. putida</i> KT2440	Experimental proof that threonine biosynthesis is required for growth on minimal medium in this exact strain
11368768	Integrated threonine-pathway enzyme kinetics in <i>E. coli</i>	Establishes the five-step aspartate→threonine pathway sequence with ThrC as the terminal step
7902068	<i>E. coli</i> threonine synthase inhibition by homoserine phosphate analogues	Quantifies competitive inhibition (K _i 6 μM, 31 μM); probes active-site substrate recognition
6150934	Aspartate antimetabolites and threonine-pathway enzymes	Reconstituted pathway; identifies hydroxyhomoserine phosphate as a potent ThrC inhibitor
9748328	Allosteric activation of <i>Arabidopsis</i> threonine synthase by SAM	Contrasts plant (SAM-activated) vs. bacterial ThrC — informs expectation that <i>P. putida</i> (bacterial) ThrC is NOT SAM-regulated

The evidence is highly consistent. The primary annotation (database + domains) is independently corroborated by direct sequence orthology (34% full-length identity), by conserved-motif mapping of the catalytic Lys112, by strain-specific genetics showing conditional essentiality, and by the well-characterized enzymology and inhibitor pharmacology of the ThrC family. No conflicting evidence was found — no literature was encountered describing a different function for a gene with this locus tag, and the organism, domains, and family are all mutually consistent.

One nuance worth flagging: plant threonine synthases are strongly allosterically activated by S-adenosylmethionine (SAM), producing an ~8-fold increase in catalytic rate and 25-fold decrease in K_m (PMID: 9748328), whereas bacterial threonine synthases (the class to which PP_0662 belongs) generally lack this regulation. This distinction is relevant to expectations about the *P. putida* enzyme's regulatory behavior.

Limitations and Knowledge Gaps

1. **No direct enzymology on the *P. putida* protein.** All catalytic parameters (k_{cat} , K_m , inhibitor constants) and structural details are inferred from orthologs (*E. coli*, *T. thermophilus*, *M. tuberculosis*, *B. subtilis*), not measured for Q88Q36 itself. The 34% identity to *E. coli* ThrC is strong but does not guarantee identical kinetic parameters.
 2. **No experimental structure of PP_0662.** The fold, domain-closure behavior, and Lys112 assignment are homology-based inferences; no crystal or cryo-EM structure of the *P. putida* enzyme exists in this analysis.
 3. **Localization is inferred, not demonstrated.** Cytoplasmic localization is a strong inference from the pathway and from the absence of targeting signals, but has not been experimentally verified for PP_0662.
 4. **Regulation is uncharacterized in *P. putida*.** Whether PP_0662 is subject to feedback regulation, allosteric effectors, or transcriptional control by the neighboring AsnC/Lrp-family regulator (PP_0663) is unknown. Bacterial ThrCs generally are not SAM-activated, but this has not been tested here.
 5. **Conditional essentiality vs. strict essentiality.** The mutant screen shows threonine auxotrophy on minimal medium; whether PP_0662 is dispensable under threonine-supplemented or complex-medium conditions (as expected for an auxotrophy) is implied but the phenotype magnitude and any redundancy have not been quantified.
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Proposed Follow-up Experiments / Actions

1. **Recombinant enzymology.** Overexpress and purify Q88Q36, confirm PLP binding spectroscopically, and measure steady-state kinetics (k_{cat} , K_m for O-phospho-L-homoserine)

plus K_i values for APPA and L-2-amino-5-phosphonovaleric acid to benchmark against *E. coli*.

2. **Site-directed mutagenesis of Lys112.** Generate a K112A variant and confirm loss of activity and loss of PLP Schiff-base absorbance, directly validating the predicted catalytic residue.
3. **Complementation and knockout.** Delete PP_0662 and confirm threonine auxotrophy; complement in trans to demonstrate that the phenotype is attributable to this locus.
4. **Structure determination.** Solve the crystal structure (apo and with a phosphonate substrate analogue such as APPA) to verify the fold-type II architecture and capture domain closure; alternatively, generate and validate an AlphaFold model.
5. **Regulatory test.** Assay whether SAM, threonine, or other pathway intermediates modulate activity, and test whether the adjacent AsnC/Lrp regulator (PP_0663) controls thrC expression — clarifying regulation at this branch point.
6. **Metabolic flux analysis.** Quantify partitioning of O-phospho-L-homoserine between the threonine (ThrC) and methionine branches under different growth conditions to define PP_0662's role in flux control.

Conclusion

PP_0662 (Q88Q36) is confidently and unambiguously identified as **threonine synthase (ThrC, EC 4.2.3.1)** of *Pseudomonas putida* KT2440 — a soluble, cytoplasmic, PLP-dependent (fold-type II / TrpB-PALP) enzyme catalyzing the final β,γ -replacement step of L-threonine biosynthesis: $\text{O-phospho-L-homoserine} + \text{H}_2\text{O} \rightarrow \text{L-threonine} + \text{Pi}$. The catalytic PLP lysine is Lys112. The enzyme completes the aspartate-derived pathway (conditionally essential in this strain), operates at the O-phosphohomoserine branch point shared with methionine biosynthesis, and — being absent in mammals — constitutes a validated selective antimicrobial/herbicide target inhibitable by phosphonate substrate analogues.